Subject Description Form

Subject Code	AAE5304
Subject Title	Safety, Reliability and Airworthiness Requirement for Low-altitude Aerial Vehicle
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject aims at to equip students with the knowledge and skills necessary to navigate the complex lower airspace ecosystem. Students will gain a comprehensive understanding of the regulatory framework, stakeholder interests, reliablity analysis and safety considerations governing lower airspace operations. The programme will explore into legal and regulatory frameworks, stakeholder interests, reliablity analysis ,safety assessment methodologies for UAVs and other low-altitude aircraft, incident and accident investigation principles, and aerial vehicle certification requirements and airworthiness considerations. Upon completion, students will be prepared to contribute meaningfully to the development and safe integration of low-altitude aviation technologies.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Analyse the legal and regulatory framework governing lower airspace operations, identifying key challenges and opportunities for innovation. This includes understanding the role of various stakeholders and the impact of emerging technologies on airspace management; b. Apply reliability analysis techniques to assess and improve the reliability of low-altitude aerial vehicle systems and components c. Design and implement effective safety assessment protocols for unmanned aerial vehicles (UAVs) and other low-altitude aircraft, ensuring compliance with regulatory standards and mitigating potential risks. This includes applying risk assessment methodologies and developing safety procedures to ensure safe and responsible operations; and d. Apply investigative techniques, analysing data, and communicating findings to relevant stakeholders.
Subject Synopsis/ Indicative Syllabus	**Lower Airspace Regulation and Policy** - introduce the lower airspace environment, explore legal and regulatory frameworks governing operations, delve into international standards and best practices, examine airspace management and access, and analyse policy and economic considerations. **Stakeholder Interests and Collaboration** - focus on analysing the diverse interests of stakeholders involved in lower airspace activities, including operators, manufacturers, regulators, and communities. Students will explore industry perspectives, regulatory perspectives, community concerns, and collaboration strategies for successful integration. Certification and Manufacturing in UAV/eVTOL Development -introduce overview of manufacturing techniques of low- altitude aerial vehicle, understand regulatory frameworks and type certification requirements Reliability and Maintainability Analysis – introduce Reliability analysis

techniques: Failure Mode and Effects Analysis (FMEA), Weibull Analysis and Reliability Block Diagram approach. Students will be able to determine the life behaviour of a low- altitude aerial vehicle component , calculate the failure rate of a low-altitude aerial vehicle system and provide a cost effective predictive maintenance solutions.

Safety Assessment and Risk Management - equip students with the skills to conduct thorough safety assessments for unmanned aerial vehicles (UAVs) and other low-altitude aircraft. Topics covered include safety management systems, risk assessment and mitigation, human factors in aviation safety, operational safety procedures and standards, and safety data analysis and reporting.

Incident and Accident Investigation - prepare students to effectively investigate incidents and accidents in the lower airspace. Students will learn the principles of accident investigation, data collection and analysis, human factors in accidents, safety recommendations and implementation, and review case studies and best practices.

Teaching/Learning Methodology

Lectures are used to deliver the knowledge of compliance topics to the students. Case study coupled with group presentations will be used to foster students' understanding of the subject matters.

Teaching/Learning Methodology	Outcomes			
	a	b	c	d
Lecture	✓	✓	✓	✓
Case study		✓	✓	
Group presentation		✓	✓	✓

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
		a	b	c	d
1. Examination	50%	✓	✓	✓	✓
2. Assignment	20%	✓	✓	✓	✓
3. Reports and presentation (Case Study)	30%	✓	✓		
Total	100 %				

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Overall Assessment:

0.5 x End of Subject Examination + 0.5 x Continuous Assessment

<table>
<tr><td></td><td colspan="2">Formal examinations will assess students' knowledge of relevant regulations, maintenance procedures, and basic airworthiness concepts. Students will engage in self-directed learning through case study reports, analyzing real-world scenarios related to safety incidents and reliability issues involving low-altitude aerial vehicles. This fosters critical thinking and problem-solving skills. This combination of assessment methods ensures a well-rounded learning experience, combining theoretical knowledge with practical application and real-world insights.</td></tr>
<tr><td rowspan="8">Student Study Effort Expected</td><td>Class contact:</td><td></td></tr>
<tr><td>▪ Lecture</td><td>33 Hrs.</td></tr>
<tr><td>▪ Tutorials</td><td>6 Hrs.</td></tr>
<tr><td>▪ Other student study effort:</td><td></td></tr>
<tr><td>▪ Assignments</td><td>20 Hrs.</td></tr>
<tr><td>▪ Report</td><td>60 Hrs.</td></tr>
<tr><td>Total student study effort</td><td>119 Hrs.</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>Reading List and References</td><td colspan="2">1. “Unmanned Aircraft Systems: UAVS Design, Development and Deployment”. Allan Seabridge, Ian Moir, Roy Langton, Reg Austin.
2. "Aircraft Design: A Conceptual Approach” 3rd Ed. Daniel P. Raymer.
3. Cusick, Stephen, Commercial Aviation Safety, Sixth edition. McGraw Hill Professional, 2017.
4. "Human Factors in Aviation" by Earl Wiener.
5. Kinnison, Harry A., and Tariq Siddiqui. "Aviation maintenance management." (2013).
6. Van de Poel, I. and Royakkers, L., 2023. Ethics, technology, and engineering: An introduction. John Wiley & Sons.
7. Elsayed A. Elsayed. 3rd Edition, 2021. “Reliability Engineering”. John Wiley & Sons
8. Special Condition for small-category VTOL-capable aircraft. Doc No. SC-VTOL-02, Issue 2, June 2024, EASA
9. Easy Access Rules for small category VCA (SC-VTOL + MOC) (Revision 0), Oct 2024, EASA
10. Means of Compliance with the Special Condition VTOL, MOC SC-VTOL Issue 2, May 2021, EASA
11. Guidelines on Design verification for UAS operated in the ‘specific’ category, Issue 3 26 March 2023, EASA
12. Easy Access Rules for Unmanned Aircraft Systems, July 2024, EASA
13. Guidelines for the Assessment of the Critical Area of an Unmanned Aircraft, May 2024, EASA</td></tr>
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Mar 2025